

MUSTAFA SHOUKAT

AI Software Engineer

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SPECIALIZATIONS

- AI Product Development (Web & Mobile Apps)
- Local & Agentic RAG System Development
- Generative AI SaaS Product Development
- AI System Architecture Design & Implementation
- Cloud-Native AI Deployment (AWS & Scalable Environments)
- Low-Code / No-Code AI Automation Solutions

SUMMARY

AI Software Engineer with 6 years of experience shipping production AI systems for clients in KSA, UK, and USA. Active open-source contributor to LangChain, RAGAS, and Unstructured with CI-passing PRs. Experienced with air-gapped deployments for government networks in the Gulf region.

TECHNICAL SKILLS

AI/ML Frameworks: LangChain, LangGraph, CrewAI, AutoGen, LlamaIndex, HuggingFace Transformers, PyTorch, scikit-learn

RAG & Retrieval: Hybrid retrieval (dense + BM25 + RRF), cross-encoder reranking, semantic chunking, Arabic NLP pipelines

Vector Databases: Qdrant, FAISS, Pinecone, Milvus, ChromaDB

RL & Evaluation: NLI scoring, Recall@k, MRR, NDCG, pytest (617+ tests), Gymnasium RL environments

LLM Fine-Tuning: LoRA, QLoRA, SFT

Languages & Web: Python, TypeScript, SQL, FastAPI, React, Next.js, Gradio

Infrastructure: Docker, AWS, GCP, Azure, GitHub Actions CI/CD, air-gapped deployments

Automation: n8n, Twilio, Make.com, Zapier, Flowise, GoHighLevel

Observability: LangSmith, AgentOps, MLflow

EXPERIENCE

AI Software Engineer, Navid-Wattad Digital, KSA (Remote) | 2024 - Present

- Built **BayanRAG 2.2**, production Arabic legal RAG: 5 ML models, hybrid retrieval (dense + BM25 + RRF), cross-encoder reranking, NLI-based confidence scoring, and **617+ automated tests** across 4-stage pipeline.
- Architected **reproducible, sandboxed environments** via 7-service Docker Compose: isolated per-tenant deployments, **air-gap support for government networks**, and 65%+ CI coverage.
- Built **agentic AI workflows** and internal SaaS tools with LangChain, LangGraph, and FastAPI, serving multiple business units with automated document processing and intelligent Q&A.

Generative AI Engineer, AutorecAI, UK (Remote) | 2022 - 2024

- Built production **agentic AI systems** with custom tool-use orchestration, autonomous task execution, and automated output validation for enterprise clients.
- Designed **containerized, air-gapped RAG deployments** for regulated environments: zero external dependencies, reproducible Docker setup, serving legal and compliance document Q&A.
- Deployed on **AWS and GCP** with Docker, GitHub Actions CI/CD, and automated regression testing.

Software Automation Engineer, Comet Estimating LLC, USA (Remote) | 2020 - 2022

- Built Python automation pipelines with **data validation and QA workflows**, CI/CD on Google Cloud Run.
- Developed **lead scoring system with human-in-the-loop review**, combining automated qualification with manual verification. Received **Best Performance Award**.

OPEN-SOURCE CONTRIBUTIONS

- **LangChain** (92k+ stars): Fixed malformed function dicts in tool call parsers + 16 tests (PR #36680)
- **RAGAS** (7k+ stars): 3 PRs — bug fixes, multilingual support (Arabic/CJK), backward-compatible import aliases + 62 tests
- **Unstructured** (14k+ stars): PaddleOCR language mapping fix + 6 tests (PR #4329)

PROJECTS

- **arabic-rag-bench:** Open-source Gymnasium RL environment for training LLMs to debug Arabic RAG pipelines. Deterministic Recall@5 scoring, reward-hacking defenses, 24 tests, CI (Python 3.10/3.11/3.12).
- **AI Receptionist SaaS:** WhatsApp-based AI receptionist for real estate and legal firms. Twilio + n8n + GoHighLevel integration.

EDUCATION

MS Artificial Intelligence, National University of Sciences and Technology (NUST), Pakistan (2025 - Present)

BS Computer Science, Virtual University (2023 - 2027)